Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AI-Based Real-Time Patient Safety Monitoring and Decision Support Platform for Diagnostic and Therapeutic Endoscopy			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio·Medical		Information & Communication	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Information & Systems	Remote and Home Health care Devices	U-Computing	U-Computing Platforms and Application Technologies
Concept of Proposed Technology	The proposed technology is an **AI-based real-time clinical monitoring and decision support platform** designed for **diagnostic and therapeutic endoscopy**. The system continuously collects and integrates **multimodal clinical data**, including wearable biosignals (heart rate, oxygen saturation, respiratory rate, body temperature, and activity), procedural parameters, and electronic medical records. Using advanced machine learning and time-series analysis algorithms, the platform generates **real-time risk predictions** for intra-procedural and post-endoscopic complications, including bleeding, infection, cardiopulmonary deterioration, sepsis, and unexpected clinical decompensation. Particular emphasis is placed on **sedation-related adverse events** such as hypoxia, respiratory depression, hypotension, arrhythmia, aspiration, and sedation-induced cardiopulmonary complications occurring during and immediately after endoscopy. The platform incorporates **explainable artificial intelligence functions** that provide transparent interpretation of risk factors and clinical relevance, supporting clinician understanding and trust. It is designed for deployment across **inpatient wards, outpatient endoscopy units, and post-**			

	discharge home monitoring environments, enabling continuity of patient safety management throughout the care pathway.
Rationale for Joint R&D Project	Endoscopic procedures are increasing worldwide due to population aging and expanded indications for minimally invasive diagnosis and treatment. Despite these benefits, endoscopy-related complications remain a major cause of preventable morbidity, prolonged hospitalization, and increased healthcare costs. Current monitoring practices rely largely on intermittent vital sign measurements and manual observation, which are insufficient for early detection of rapid physiological deterioration, particularly under procedural sedation. Sedation-related cardiopulmonary events continue to represent one of the leading causes of endoscopy-related mortality and medicolegal disputes, while predictive real-time monitoring solutions remain limited in routine clinical use. Joint R&D is required to bridge the gap between advanced clinical AI research and real-world medical product development. The research institute contributes clinical expertise, algorithm development, and access to real-world datasets, while the Korean SME partner provides system engineering, wearable device integration, regulatory preparation, and commercialization capabilities. Through joint development, the project enables: • Translation of AI algorithms into clinically deployable systems • Validation across multiple clinical environments, including overseas institutions • Alignment with international medical device regulatory frameworks for Software as a Medical Device (SaMD)
Key Words	1. Diagnostic and Therapeutic Endoscopy 2. Medical Artificial Intelligence 3. Real-Time Clinical Monitoring 4. Sedation-Related Adverse Events 5. Multimodal Clinical Data

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3-4)	End of the Project	Level (7-8)
R&D Objectives	The primary objective of this project is to develop a **clinically validated AI-based real-time monitoring and decision support platform** capable of predicting intra-procedural and post-endoscopic complications and supporting proactive clinical intervention during diagnostic and therapeutic endoscopy. Specific objectives include: 1. Development of advanced AI algorithms for early prediction of sedation-related cardiopulmonary complications, bleeding, infection, and sepsis 2. Construction of a real-time data integration platform linking wearable biosignal devices with hospital EMR and endoscopy information systems 3. Development of clinician-oriented dashboards and early warning alert systems 4. Implementation of explainable AI functions for transparent interpretation of risk factors and clinical relevance 5. Execution of multicenter clinical pilot studies, including overseas medical institutions, and establishment of a technical foundation for SaMD regulatory approval			
R&D Target Specifications	• Complication prediction performance: **AUC ≥ 0.85** • Early warning time before clinical diagnosis: **≥ 6–12 hours** • False alarm rate: **≤ 20%** • Continuous monitoring uptime: **≥ 95%** • System response latency: **≤ 5 seconds per prediction cycle** • Data integration success rate (wearable + EMR): **≥ 98%** • Alert precision (PPV): **≥ 70%** • Clinical workflow usability: **Clinician satisfaction ≥ 80%** • Regulatory readiness: **Pre-submission documents for MFDS/FDA prepared**			
R&D Plan	**Phase 1 (Year 1): AI Algorithm Development and Data Preparation** • Collection and preprocessing of multimodal clinical datasets related to endoscopic procedures • Development of deep learning and time-series prediction models for sedation-related cardiopulmonary events, bleeding, and infection • Feature engineering based on physiological signals and procedural parameters • Internal validation using retrospective endoscopy cohort data **Phase 2 (Year 2): System Integration and Prototype Development** • Integration of wearable biosignal devices with hospital information systems and endoscopy platforms • Development of real-time data pipelines and cloud-based AI inference engines • Construction of clinician dashboards and early warning alert interfaces • Performance testing in simulated clinical environments and single-center pilot studies **Phase 3 (Year 3): Clinical Validation and Global Pilot** • Execution of prospective multicenter clinical pilot studies, including overseas endoscopy centers • Model refinement based on real-world monitoring data • Evaluation of clinical effectiveness, safety, and usability			

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	• Preparation of regulatory documentation for SaMD and development of a global commercialization roadmap

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Provide high-quality endoscopy-related clinical datasets, including procedural data, patient outcomes, and multimodal physiological signals • Design clinical research protocols and manage ethical approvals for retrospective and prospective studies • Develop and optimize AI algorithms for early detection and risk prediction of endoscopic complications • Perform feature engineering, model training, and performance evaluation using real-world cohorts • Conduct internal and external validation and generate scientific evidence for clinical safety and effectiveness • Provide domain expertise in endoscopic workflows, sedation-related risk mechanisms, and clinical decision-making • Coordinate international collaborative research activities and lead scientific dissemination
Expected R&R of Korean SME Partner	• Provide wearable biosignal monitoring hardware and system software with clinical reliability • Build cloud-based AI infrastructure for real-time biosignal transmission and inference • Integrate endoscopy systems, wearable monitoring systems, and hospital EMR/HER • Develop clinician-oriented user interfaces and alert dashboards • Conduct system-level testing, performance verification, and operational validation • Prepare regulatory approval processes for SaMD in global markets (MFDS, CE, FDA) • Establish productization strategies, intellectual property management, and commercialization plans • Manage technical collaboration with overseas healthcare AI and industrial partners

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Outcomes** • Development of a clinically validated AI-based real-time monitoring and decision support platform for endoscopy • Establishment of advanced AI algorithms for early prediction of sedation-related cardiopulmonary complications, bleeding, infection, and sepsis • Development of multimodal data integration technologies combining wearable biosignals, endoscopy systems, and hospital EMR • Creation of explainable AI-based clinical decision support tools aligned with endoscopic workflows • Accumulation of international clinical evidence supporting safety, reliability, and effectiveness • Formation of a core technology platform extensible to other minimally invasive procedures **Commercialization Potential** • Preparation for regulatory approval and certification in major global markets • Development of scalable SaMD products deployable in hospitals, outpatient endoscopy centers, and post-discharge home monitoring services

	• Creation of digital healthcare business models, including subscription-based AI monitoring services and integrated clinical platforms • Establishment of strategic partnerships with global medical device and healthcare AI enterprises • Strengthening of intellectual property portfolios and technology transfer opportunities **Industrial and Economic Impact** • Reduction of preventable complications, mortality, and healthcare costs through early detection and proactive clinical intervention • Improvement of workflow efficiency and patient safety in endoscopy units • Enhancement of international competitiveness of Korean SMEs in medical AI and digital health sectors • Expansion of export opportunities and creation of high-value digital healthcare markets • Contribution to national strategic goals in Advanced Bio and Artificial Intelligence industries

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